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Roll No.....

Degree: B.Tech. Semester: SUMMER - Course work

MID-SEMESTER EXAMINATION, June 2026

Course Title: *Mathematics III*

Course Code: ICMTC04

Time: 1.5 Hours

Max. Marks: 25

Note: - Attempt all questions in given. Missing data/ information (if any), maybe suitably assumed & mentioned in the answer.

Q. No.	Question	Marks	CO
Q 1			
1(a)	Show that the function $u(x, y) = x^4 - 6x^2y^2 + y^4$ is harmonic.	2.5	CO1
1(b)	Find the analytic function $f(z)$ whose real part is $u(x, y) = e^x(x \cos y - y \sin y)$ and $f(1) = e$.	2.5	CO1
Q 2			
2(a)	Evaluate the following integral using Cauchy Integral formula: $\oint_C \frac{\cos(\pi z)}{z^2(z-1)} dz$ where C is the circle $ z = 2$.	2.5	CO1
2(b)	Obtain the Taylor series expansion of the function $f(z) = \frac{z}{z^2 - 3z + 2}$ about the point $z = 0$.	2.5	CO1
Q 3			
3(a)	Find the residue at each pole of the function $f(z) = \frac{\exp z}{z^2 + \pi^2}$.	2.5	CO1
3(b)	If $\vec{r} = xi + yj + zk$ and $r = \vec{r} $, then prove that $\text{curl}(\text{grad } r^m) = \vec{0}$.	2.5	CO2
Q 4			
4(a)	Find the directional derivative of the function $\phi(x, y, z) = x^2 - y^2 + 2z^2$ at the point $P(1, 2, 3)$ in the direction of the line PQ , where Q has coordinates $(5, 0, 4)$.	2.5	CO2

Q. No.	Question	Marks	CO
4(b)	A fluid motion is given by $\vec{v} = (y + z)\mathbf{i} + (z + x)\mathbf{j} + (x + y)\mathbf{k}$. Is this motion irrotational? If so, find the scalar potential.	2.5	CO2
Q 5			
5(a)	A vector field is given by $\vec{F} = x^2\mathbf{i} - xy\mathbf{j}$. Find the line integral $\int \vec{F} \cdot d\vec{r}$ from the origin $O(0, 0)$ to the point $P(1, 1)$ along the x -axis from O to $A(1, 0)$ and then from $A(1, 0)$ to $P(1, 1)$.	2.5	CO2
5(b)	Using Stokes theorem, find $\oint_C [(x + y)dx + (2x - z)dy + (y + z)dz]$, where C is the boundary of the triangle with vertices $(2, 0, 0)$, $(0, 3, 0)$, and $(0, 0, 6)$.	2.5	CO2